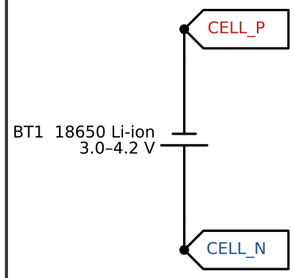
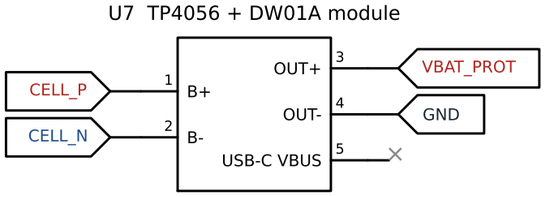
PROPOSED DESIGN — NOT YET BUILT OR TESTED. Net labels of the same name are connected. Module outlines are breakout boards; pin numbers are this design's own connector order.

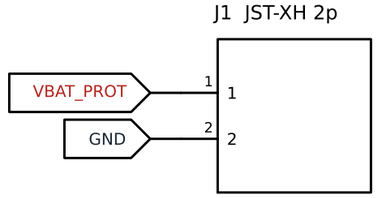
Battery assembly (cell on face B; U7 on face A below the carrier)

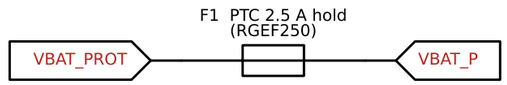




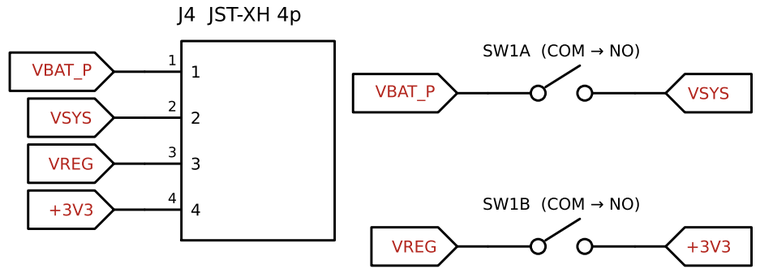
U7: DW01A over-charge / over-discharge / over-current protection with FS8205A switch.
Its TP4056 USB-C charge input is used only for supervised bench charging with the
shell removed; preferred charging is in an external 18650 charger.

Carrier power entry and fuse



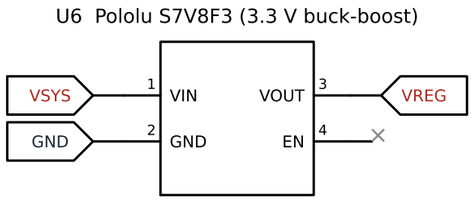


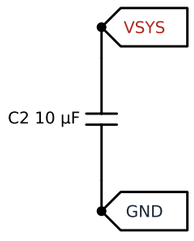
Master switch SW1 (bottom cap) via J4

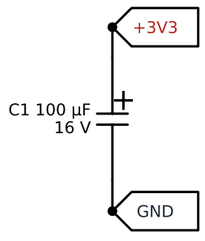


DPDT: pole A connects the protected cell to VSYS;
pole B connects the regulator output to the +3V3 rail.
OFF isolates the regulator so the ESP32 USB port can
power the board for programming (one supply at a time).
NC contacts of both poles are unused.

3.3 V regulator and bulk capacitance

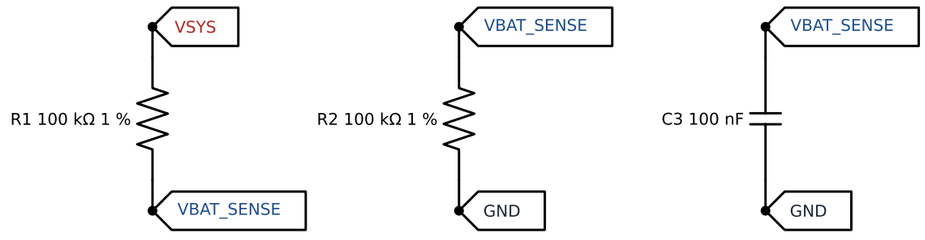






U6 EN left open (internal pull-up enables the regulator; confirm on the Pololu page).
C1 sits at the radio connector end of the +3V3 rail to supply transmit peaks.

Battery-voltage sense (ADC1_CH6)



VBAT_SENSE = VSYS / 2 (2.1 V at 4.2 V).
Divider draws 21 µA at 4.2 V.
Read with 11 dB attenuation on IO34 (ADC1),
calibrated against a multimeter.

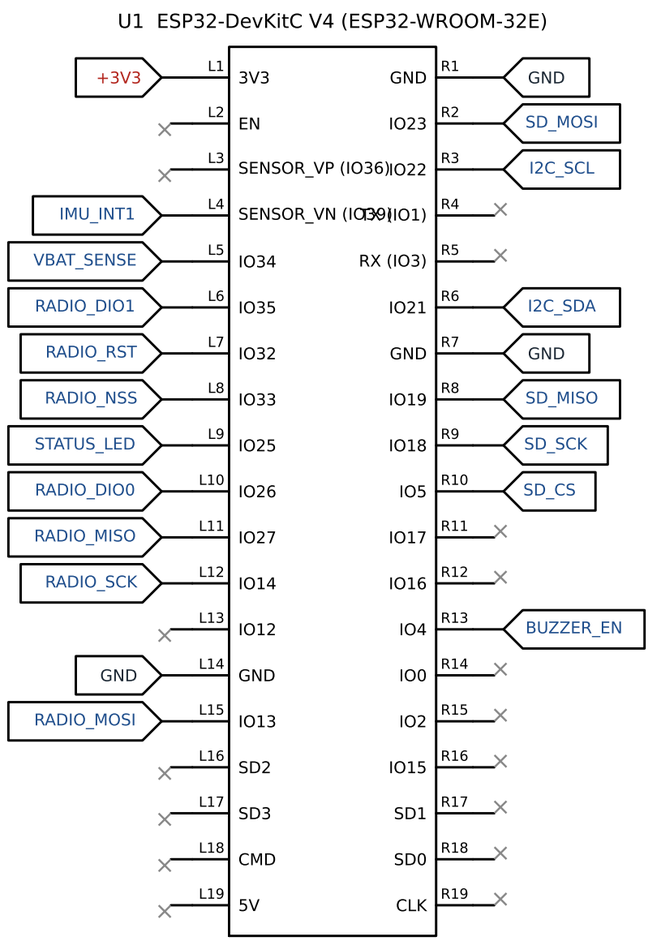
STRATOS — Power and regulation

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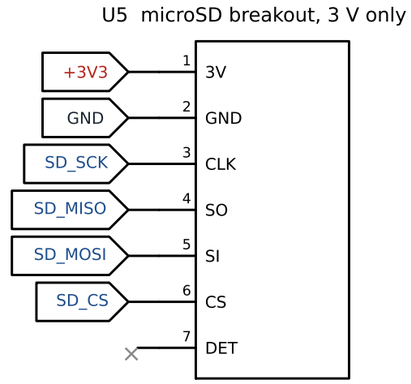
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PROPOSED DESIGN — NOT YET BUILT OR TESTED. Net labels of the same name are connected. Module outlines are breakout boards; pin numbers are this design's own connector order.

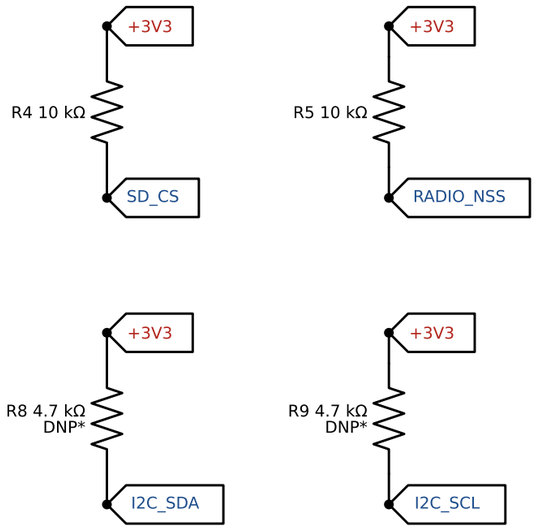
U1 ESP32-DevKitC V4 on two 1×19 female headers (left = 3V3 end header, right = GND end header)



microSD (VSPI)

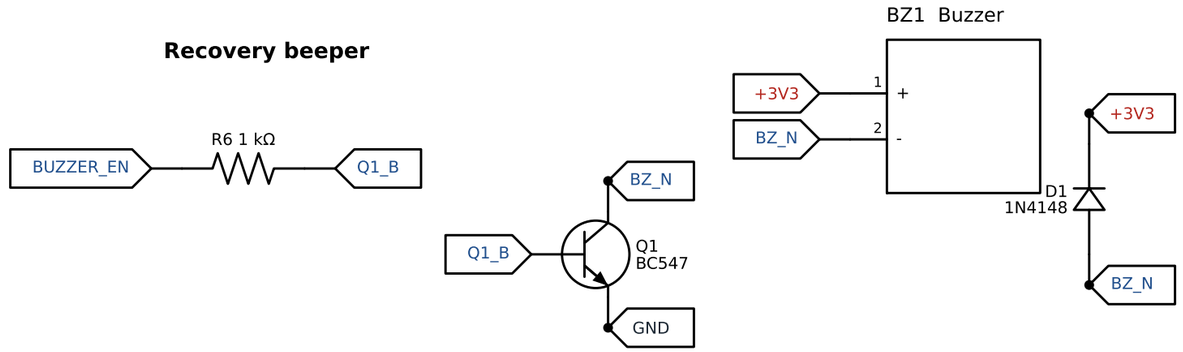


Boot-safe pull-ups

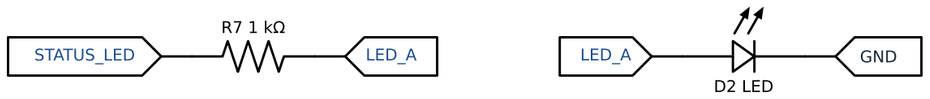


R4 holds IO5 (strapping pin) high at boot and keeps the card deselected; R5 keeps the radio deselected. *DNP = do not populate: fit only if the breakouts lack I2C pull-ups.

Recovery beeper



Status LED



× = not connected. IO12 and IO15 (strapping), IO0/IO2 (boot), SD0–SD3/CMD/CLK (internal flash) are left unused. IO16/IO17 reserved for an optional GNSS receiver (UART2), not fitted in the baseline. TX/RX = USB programming only.

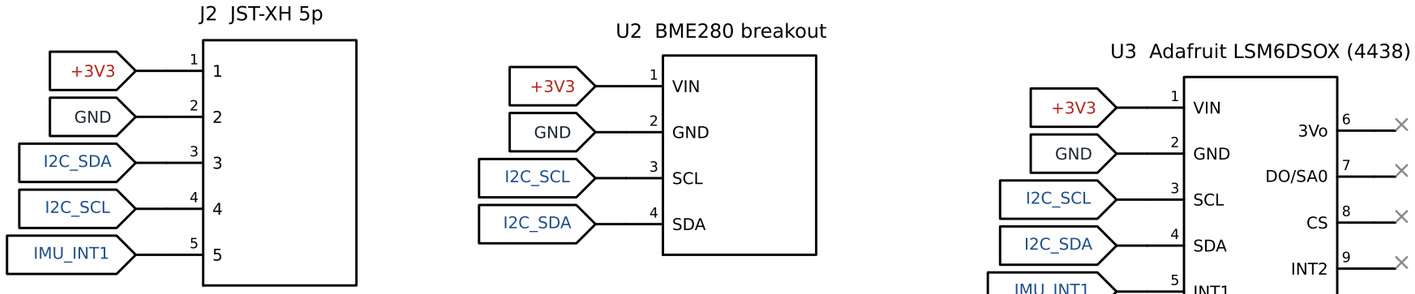
STRATOS — Controller, storage, indicators

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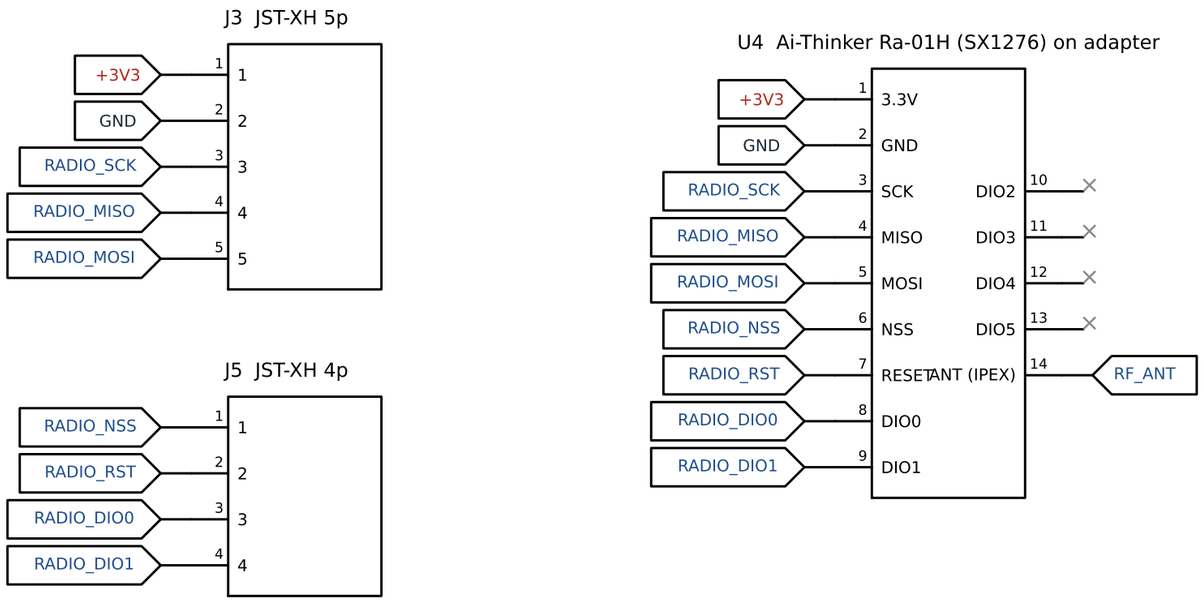
PROPOSED DESIGN — NOT YET BUILT OR TESTED. Net labels of the same name are connected. Module outlines are breakout boards; pin numbers are this design's own connector order.

Sensor bus (J2 on carrier → U2, U3 on face B)



I2C address: BME280 0x76 or 0x77 (set by the breakout's SDO strap — read the actual board);
LSM6DSOX 0x6A (Adafruit default). Bus at 400 kHz; total pull-up resistance to be measured (R8/R9 DNP).
INT1 = IMU data-ready, wired to input-only IO39 (no internal pull needed: push-pull output).

Radio (J3 SPI + J5 control on carrier → U4 on face B)



Radio configured for GFSK, 866.0 MHz, 38.4 kb/s, ±40 kHz,
+14 dBm on PA_BOOST (see dossier §4.6 for limits).
C1 (sheet 1) is placed at the J3 end of the +3V3 rail.
DIO2–DIO5 unused. Keep the RF feed short; antenna at
least 30 mm from the cell and ESP32 module.

STRATOS — Sensors (I2C) and radio (HSPI)

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